William Moreira Heck

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Education

University of South Florida - Judy Genshaft Honors College | Tampa, FL — **Expected Spring 2030**

Bachelor of Science in Mechanical Engineering
Scholarship: Florida Academic Scholars Award (Bright Futures)

Horizon High School | Winter Garden, FL — **May 2026**

High School Diploma; GPA: 3.1/4.0

Engineering & Projects

IEEE USF Exoskeleton Project - Mechanical Team | Mechanical Integration, CAD, Human Interface — **2026 - Present**

- Support mechanical subsystem development for a unilateral lower-limb exoskeleton with powered hip and knee joints and a passive ankle architecture.
- Contribute to CAD assembly work, actuator mounting interfaces, structural links, cuffs and straps, joint-alignment features, fabrication drawings, BOM development, fit checks, and assembly documentation.
- Proposed a computer-vision-controlled robotic arm concept that can mimic human arm motion and integrate with the team's hand work; the design is currently being developed around an Abenics spherical gear shoulder mechanism.
- Help with early CAD development for the robotic arm, including joint layout, component placement, packaging, and mechanical design decisions for the spherical-gear shoulder.

IEEE USF SoutheastCon Robotics Team | Competition Robot Design & Build — **2026 - Present**

- Work with a student engineering team to design and build a robot for the IEEE SoutheastCon robotics competition while operating within the team's competition requirements and budget constraints.
- Research drivetrain and motor options by comparing torque, speed, voltage, encoder feedback, cost, and suitability for the robot's expected tasks.
- Contribute to mechanical concept development and CAD discussions for mechanisms used to interact with and collect competition objects.

Independent Automotive Restoration | Diagnostics, Repair, Parts Sourcing, Resale — **2024 - 2025**

- Sourced vehicles through dealership auctions with a small team, assessed repair needs, planned work, purchased parts, tracked costs, and prepared completed vehicles for resale.
- Restored a 2014 Subaru BRZ through body repair, component replacement, cleaning and paint preparation, reassembly, and troubleshooting of manual-transmission shifting and suspected synchronizer-related issues with experienced mechanic support.
- Helped return a non-running 2013 Dodge Dart purchased for $500 to drivable condition through transmission and shift-linkage troubleshooting, component repair or replacement, and exterior restoration.

Experience

Server | White Rabbit | Winter Garden, FL — **2023 - 2026**

- Provided fast, accurate service in a high-volume restaurant, handled customer issues in real time, and coordinated with kitchen staff and management during peak periods.
- Balanced consistent 15-20+ hour work weeks alongside high school academics, building reliability, communication, teamwork, and time-management skills.

Skills

Technical: SolidWorks/CAD fundamentals, mechanical integration, BOMs, fabrication drawings, subsystem documentation
Robotics: Motor selection research, drivetrain concepts, actuator integration, encoder fundamentals, mechanism design, computer-vision project collaboration
Mechanical: Automotive troubleshooting, disassembly/reassembly, component replacement, hand and power tools, basic fabrication, fit checks
Project: Parts sourcing, cost tracking, repair planning, team collaboration, design research, customer communication
Soft Skills: Communication, teamwork, problem solving, time management, adaptability, reliability, customer service
Languages: English and Portuguese

Relevant Coursework & Interests

Coursework: Calculus I, General Chemistry I & Lab, Foundations of Engineering, Honors coursework
Engineering Interests: Robotics, aerospace, automotive systems, motorsports engineering, mechanical design, and controls